ISCC
International Sustainability & Carbon Certification
Guidance



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PLUS

ISCC PLUS GHG Add-on Document

Version 1.0

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Table of Abbreviations

Abbreviation	Name
CB	Certification Body
DDV	Disaggregated Default Values
EAC	Energy Attribute Certificate
EF	Emission Factor
GHG	Greenhouse Gas
GoO	Guarantees of Origin
iLUC	Indirect Land Use Change
ISAE	International Standard on Assurance
ISCC	International Sustainability and Carbon Certification
ISO	International Standardization Organization
LCA	Life Cycle Assessment
LUC	Land Use Change
LULUCF	Land Use, Land Use Change, and Forestry
RED	Renewable Energy Directive
RFNBO	Renewable Fuels of Non-Biological Origin
SD	Sustainability Declaration
TDV	Total Default Values
UCO	Used Cooking Oil

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1. Introduction

The purpose of the document "ISCC PLUS GHG Add-on Document" is to explain the use of the voluntary GHG add-on under ISCC PLUS for stating greenhouse gas (GHG) emissions values along the supply chain and to provide the methodology, rules and guidelines for calculating and verifying GHG emissions and emissions reduction. Within ISCC PLUS, the verification of GHG emissions can be added by applying the 205-01 "GHG Emissions" add-on.

The ISCC requirements regarding GHG emissions apply to all relevant supply chain elements from raw material production to the distribution of the final product, including cultivation or extraction, all processing steps, and the transport and distribution of intermediate and final products.

The system users can follow two different methodologies under ISCC PLUS GHG add-on. The two different methodologies that can be applied are:

1. Methodology introduced in the System Document *ISCC EU 205 – Greenhouse Gas Emissions*

2. Methodology introduced in the System Document *ISCC EU 205-1 – Renewable Fuels of Non-Biological Origin (RFNBO) and Recycled Carbon Fuels (RCF) Greenhouse Gas Emissions*

If the add-on is applied, the System Documents *(ISCC EU 205 or ISCC EU 205-1)* apply equally for ISCC EU and ISCC PLUS, with some differing requirements under ISCC PLUS which are described in the following sub-chapters.

Under ISCC PLUS, it is possible to verify the calculated GHG emissions based on the methodologies described in this document, ISCC EU 205 and ISCC EU 205-1 documents. Any claims regarding GHG Emission reductions based on the calculations on the above-mentioned methodologies are beyond the scope of ISCC PLUS GHG Add-on.

In any case the GHG add-on must be implemented in the entire supply chain up to the entity claiming a value on the outgoing product.

It should be noted that GHG Add-on cannot be applied for Mass Balancing options Free Attribution Approach and Fuel Use Excluded Attribution Approach under Credit Method. As an exception for Free attribution approach, GHG Add-on can only be applied when complete reflection of input characteristics is reflected. When compensation of input characteristics is applied (either by system user or on received certified input materials), GHG add-on cannot be applied. For more information on reflection of input characteristics, please refer to *System Document 203-2 Chain of Custody.*

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In case system users have conducted a LCA based on ISO 14040/44 or ISO 14067 for product carbon footprint (PCF) calculation which differs from the ISCC methodology described in the System Documents ISCC EU 205 – Greenhouse Gas Emissions, ISCC EU 205-1 – Renewable Fuels of Non-Biological Origin (RFNBO) and Recycled Carbon Fuels (RCF) Greenhouse Gas Emissions, and in this document, the calculated value needs to be communicated separately and cannot be used to replace a GHG calculation under the 205-01 “GHG Emissions” add-on.

Next to the application of the here described 205-01 “GHG Emissions” add-on, ISCC offers the possibility to certify PCFs of ISCC PLUS certified products under the ISCC Carbon Footprint Certification (CFC) scheme1. ISCC PLUS strongly recommends the 205-01 “GHG Emissions” add-on to be applied for fuel products. For any other products with limited applicability of the 205-01 “GHG Emissions” add-on, please refer to ISCC CFC.

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[1] *The current version of ISCC CFC V1.4. allows the certification of PCFs under CFC for ISCC PLUS certified products from the PLUS raw material category “bio”. The combination for all other raw material categories is under development and can only be conducted in form of pilot projects under involvement of ISCC.*

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2. GHG Add-on following the ISCC EU 205 – Greenhouse Gas Emissions

ISCC PLUS system users can freely choose to apply the voluntary GHG add-on. When choosing to disclose the GHG emissions information from the relevant supply chain and production of product, the total GHG emissions value must be made available. Please refer to System Document *ISCC EU – 205 Greenhouse Gas Emissions* for further information regarding requirements for an individual GHG emissions calculation and provision of GHG information. The total GHG emissions value should be displayed in terms of either kg CO_2eq/dry-ton (for raw materials and intermediate products) or g CO_2eq/MJ (for final fuels).

System users have three different options for providing GHG emissions values under the ISCC PLUS GHG add-on as listed below:

1. Total Default Values (TDV)
2. Actual Values (individually calculated values)
3. Disaggregated Default Values (DDV) which allows a combination of default values and actual values.

The total GHG emissions value should be calculated following the formula below:

$$E = e_{ec} + e_l + e_p + e_{td} + e_u - e_{sca} - e_{ccs} - e_{ccr}$$

E = total GHG emissions value

e_{ec} = emissions from extraction or cultivation of raw materials

e_l = annualized emissions from carbon stock changes caused by land-use change

e_p = emissions from processing

e_{td} = emissions from transport and distribution

e_u = emissions from the fuel in use

e_{sca} = emissions savings from soil carbon accumulation via improved agricultural management

e_{ccs} = emissions savings from CO_2 capture and geological storage

e_{ccr} = emissions savings from CO_2 capture and replacement

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Apart from abovementioned elements for the calculation of total GHG emissions value, RED III also provides the option for a GHG bonus (e_b) if the degraded land (including land that was formerly in agricultural use) is restored and if the biomass is obtained from the restored degraded land. When the land is either significantly salinated or presented significantly low organic matter content and has been severely eroded, is considered severely degraded. The bonus is thus applicable in case of conversion of the land to agricultural use that increases carbon stocks. The bonus can only be applied and attributed if the evidence is provided (refer to the System Document *ISCC EU 205 – Greenhouse Gas Emissions*).

TDVs and DDVs are provided in RED III in Annex V and Annex VI. DDVs are available for cultivation (e_{ec}), processing (e_p), and transport & distribution (e_{td}). Using these values allows combining DDVs with actual data from individual GHG calculations.

In case TDVs in accordance with RED III have been used for demonstrating the GHG impact of a product, no actual figures for the GHG values can be reported. However, it must be indicated on the Sustainability Declaration (SD): "**Total default values according to RED III applied"** .

Similarly, when using DDVs for one or more elements of the calculation formula, this information must be indicated on the outgoing SD as well: "**Disaggregated default values according to RED III applied"**. Please refer to the System Document *ISCC EU 205 – Greenhouse Gas Emissions* for detailed restrictions on the use of default values.

Actual values are calculated based on RED III methodology and must always be conducted at the point in the supply chain where they originate. It is not possible to calculate actual values retrospectively for elements of the upstream supply chain. An individual GHG calculation for raw materials and intermediate products must always be indicated in terms of kg CO_2eq/dry-ton. Mode of transport and transport distance (in km) should also be provided by system users in case default values are not considered for the GHG emissions calculation. Values from the specific "NUTS2 region" report can be used as an alternative to DDVs or actual values. In case NUTS2 values are used, system users should provide this information in the SD Annex, in the "product(s) related information" field.

2.1 Specific Requirements for Elements of the Supply Chain

If the system user chooses to apply the voluntary ISCC PLUS GHG add-on, there are specific requirements depending on the scope of certificate. In this subchapter, the responsibilities for each element of the supply chain are described.

2.1.1 Farms or Plantations/Forest Sourcing Areas

When applicable, farms or plantations are responsible to provide emissions values for extraction and cultivation (e_{ec}). In case e_l, e_{sca} and e_b are applied,

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individual calculations are required. The central office is responsible for the calculation of the GHG emissions of a farmers group, if they are certified under a group approach.

2.1.2 First Gathering Point

The first gathering point is responsible for determining and documenting GHG emissions for the extraction and cultivation (e_{ec}) of the incoming biomass and for verifying whether specific options to state greenhouse gas emissions (for example, disaggregated default value for cultivation or NUTS2 values) can be applied.

2.1.3 Point of Origin for Waste and Residues

At the point of origin, GHG emissions of waste and residues are considered to be zero for the waste generation. Further emissions associated to waste transportation and treatment must be considered.

2.1.4 Processing Unit

If the processing unit produces final biofuels, bioliquids or biomass fuels, the following information should be added to the Sustainability Declaration[2]:

- Information on when the fuel producer started operation[3]
- Country of fuel production
- Energy content of the fuel (in MJ)[4]
- In case of biofuel, bioliquid or biomass production, GHG emissions of the biofuel, bioliquid or biomass fuel in g CO_2eq/MJ using the correct Feedstock Factor and Allocation Factor.
- e_p: GHG emissions from processing (applicable for intermediate and final processing units).
- In case of biomass fuels, further emissions from diffuse methane, gas losses occurring from transport, emissions during the storage of the digestate and DDV for compression at filling station (4.6 gCO_2/MJ biomethane) must be added according to RED III, Annex VI.
- For biomass fuels used in the production of electricity or used for heating and cooling, GHG emissions from final energy commodity (EC_h or EC_{el}) should be calculated.

[2] *The mandatory information for the sustainability characteristics are describes in the System Document ISCC PLUS 203-1 Traceability*
[3] *The installation shall be considered to be in operation once the physical production of fuel has started (i.e. once the production of fuels, including biofuels, biogas or bioliquids has started).*
[4] *For the calculation of the energy content the conversion factors of Annex III RED II must be used.*

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- In case of biogas/biomethane, the S_n factor formula (Part B of the Annex VI, RED II) should be used to calculate total GHG emissions[5]. Also, $e_{td,prod}$ needs to be provided in addition to $e_{td,feestock}$ for biogas/biomethane.

- For Liquefaction of Biomethane (Bio-LNG), liquefaction emissions and losses must be accounted for.

- e_{ccs}: GHG emissions savings from carbon capture and geological storage. Only applicable for CO_2 directly related to the extraction, transport, processing & distribution of fuel. Also, e_{ccs} can only be considered if it has not been accounted for in e_p

- e_{ccr}: GHG emissions savings from carbon capture and replacement. It is limited to the emissions avoided through capture of CO_2 originating from biomass. Also, only applicable for CO_2 used to replace fossil-derived CO_2

- e_u (Emissions from the fuel in use): Emissions of non-CO_2 greenhouse gases (CH_4 and N_2O) need to be included for bioliquids and biomass fuels. DDV can also be used for e_u

- e_{td}: Emissions from transport and distribution. This value should also include transport to depot and filling station as well as emissions at filling stations. Final processing unit should include e_{td} for both upstream and downstream transport

- e_l (annualised emissions from carbon stock changes caused by land-use change): Emissions from land-use change between five land categories to cropland or perennial cropland taking place shall be considered here. A change of management activities, cropland structure, tillage practice or manure input practices cannot be considered as land-use change. If the bonus (e_B= 29 g CO_2eq/MJ) for severely degraded land restored was applied, it can be utilised in the calculation of e_l (refer to the System Document *ISCC EU 205 – Greenhouse Gas Emissions)*.

- e_{sca} (Emissions savings from soil carbon accumulation via improved agricultural management): Emissions savings can be utilised due to carbon accumulation in soil driven by the adoption of improved agricultural management (which do not risk to negatively affect biodiversity).If e_{sca} is calculated, the methodology must be made available. The annual caps of the claim shall follow the requirements

[5] ***Sn*** *is share of feedstock n representing the proportion of each specific raw material entering the digester over a given year. This value is used for calculating overall lifecycle emissions (E) by summing weighted emissions from each feedstock. It serves as a variable within the overall formula, connecting emissions from individual feedstocks—such as cultivation, transport, and land-use change—to the carbon intensity of the final biogas product.*

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laid out in System Document *ISCC EU 205 – Greenhouse Gas Emissions*. A bonus of 45 g CO_2eq/MJ shall be attributed for improved agricultural and manure management in the case animal manure is used as a substrate to produce biogas and biomethane (refer to the System Document *ISCC EU 205 – Greenhouse Gas Emissions*).

2.1.5 Paper Trader and Trader with Storage

Paper traders do not calculate emissions from transport but only forward information on outgoing sustainability declarations as received. Trader with storage do not calculate emissions, however, only provide information for transport distance(s) and transport type(s) towards their storage site on the outgoing sustainability declaration. The next element in the supply chain (e.g. processing unit) after the trader with storage bears the responsibility for emissions calculation from upstream transport activities.

2.2 Deviations with respect to Emission Factors

Within ISCC PLUS, emission factors (EF) of input materials can be individually calculated or come from official sources like the Annex IX of the IR 2022/996, Renewable Energy Directive (RED) or Annex I of the System Document *ISCC EU – 205 Greenhouse Gas Emissions*. Furthermore, values based on LCA databases such as ECOINVENT or official statistical databases from government bodies or scientifically peer-reviewed literature can be used, if applicable. All data gathered from databases or literature shall be based on the most recent available versions and shall also be updated over time. Recognized emissions factor sources are:

- RED III
- IR 2022/996
- ISO 14040/44 or ISO 14067.

The methodology used for individual calculations by the supplier must always be verified according to the ISO standard or alternatively the supplier must be ISCC/ISO certified so that relevant requirements can be verified during an ISCC audit.

2.3 Calculation of Regional GHG Values for Cultivation (e_{ec})

If NUTS2 values are not available in non-EU country reports or if DDVs for cultivation emissions are missing in the RED, ISCC PLUS permits third parties (e.g. companies, plantation owners, associations) to calculate averages based on local farming practices, for example, average emissions calculated for geographical areas at national level or data from a group of farms. The methodology shall follow ISCC's requirements and ISCC should be informed whenever such values are calculated. However, a submission of the GHG emissions values to the European Commission is not required.

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2.4 Calculation of Individual GHG Values for Cultivation (e_{ec})

In case of an individual GHG emissions calculation for a group of farms or plantations, the averaging of input values and GHG emissions values is accepted under ISCC PLUS.

2.5 Aggregation of Different GHG Values

It is also possible to use the highest GHG values of the least performing batch for the same kind of certified input material (material with identical sustainability characteristic), alternatively to using single values for each certified input material.

2.6 Allocation of GHG Emissions

Under ISCC PLUS, the allocation of emissions to main and co-products can be based on the energy content of both products (refer to the System Document *ISCC EU – 205 Greenhouse Gas Emissions*), but other types of allocation (e.g. based on mass) are also possible. The most suitable allocation method should always be used, e.g. if the main product is used energetically, an energetic allocation should be applied.

2.7 Life Cycle Coverage

Under ISCC PLUS, the GHG emissions calculation can either cover the whole life cycle of the product (cradle-to-grave), or only the emissions up to the factory gate (cradle-to-gate).

It must always be clearly highlighted on the Sustainability Declaration of the product if the cradle-to-grave or cradle-to-gate approach is used. If required, further information on the additional emissions to be included for the product must be provided to the recipient of the material.

2.8 Waste and Residues

Waste and residues shall be considered to have zero GHG emissions up to the point of their collection, where the waste or residue initially occurs/is generated (point of origin scope under ISCC PLUS). This is regardless of whether the materials are further processed into intermediate products before being processed into the final product.

For example, in the case of UCO, the point of origin would be the restaurants or plants producing fried products. In the case of rendered animal fats, the point of origin would be the rendering plant producing animal fat from animal by-products. If UCO or other wastes originate from households, the point of origin could be a private company or a municipality responsible for waste collection.

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2.9 Emission Factors for Electricity

In case electricity is supplied by the grid, the emission factors of the national/country electricity mix shall be used[6]. The only exceptions to use an updated emission factor for electricity is:

Option 1 – System user receives renewable electricity via contracts

If the system user has Power Purchase Agreement (PPA) combined with Energy Attribute Certificates (e.g., Guarantees of Origin (GoO) in the EU or RECs in the U.S., I-RECs) or comparable proof of no multiple counting, the system user can choose to apply the supplier specific EF. The following information shall be a part of the documentation for using supplier specific EF:

- The energy source from which the energy carrier was produced and the start and end dates of production.
- The type of energy or energy carrier (e.g., electricity, gas–including H_2, heat, or cooling).
- The identity, location, type and capacity of the installation where the energy or energy carrier was produced.
- The date on which the installation became operational (i.e., starting point of first electricity supply into grid).
- The date and country of issue and a unique identification number of the EAC. (In other words, a proof of no double counting is required).
- The proof of renewability must ensure that double counting of the renewable electricity is avoided. A respective proof of the unit using the renewable energy must be provided during the audit.

If those requirements are not fulfilled, or, in case the supplier specific EF is not available, or contracts are sold to a third party, the emission factors of the national/country electricity mix shall be used.

Option 2 – System user has a direct connection with the respective unit producing renewable electricity

In the case that an electricity production plant is connected to the grid, using the average emission value for electricity from that individual electricity production plant in the production process is permitted. It must be guaranteed that there is a direct connection between the production facility and the individual electricity production plant and it must be possible to validate the amount of electricity used with a suitable meter.

[6] The primary source for official national and regional grid emission factors.

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In case the renewable electricity produced at the respective unit is directly consumed (i.e. not supplied from the grid), an adapted EF for the type of renewable electricity may be used.

3. GHG Add-on following the ISCC EU 205-1 – Renewable Fuels of Non-Biological Origin (RFNBO) Greenhouse Gas Emissions

ISCC PLUS system users can freely choose to apply the voluntary GHG add-on. However, only the system user that handles renewable-energy-derived raw materials and/or intermediate/final products, can follow the methodology introduced in the System Document *ISCC EU 205-1 – Renewable Fuels of Non-Biological Origin (RFNBO) and Recycled Carbon Fuels (RCF) Greenhouse Gas Emissions*.

Under ISCC PLUS, RFNBOs as defined under the RED would fall under the raw material category of "renewable-energy-derived". This should also be indicated/selected in the sustainability declaration.

GHG emissions intensity of RFNBOs shall be calculated as follows:

$$E = e_i + e_p + e_{td} + e_u - e_{ccs}$$

E = total emissions from the use of the fuel

$e_i = e_{i\,elastic} + e_{i\,rigid} - e_{i\,ex-use}\ (emissions\ from\ supply\ of\ inputs)$

$e_{i\,elastic}$ = emissions from elastic inputs

$e_{i\,rigid}$ = emissions from rigid inputs

$e_{ex\text{-}use}$ = emissions from inputs' existing use or fate

e_p = emissions from processing

e_{td} = emissions from transport and distribution

e_u = emissions from fuel in use

e_{ccs} = emissions savings from CO_2 capture and geological storage

The GHG emissions of RFNBOs shall be expressed in terms of gCO_2eq/MJ fuel. In case a fuel is a mix of RFNBO and other fuel types, all fuel types shall be considered to have the same emissions intensity. In case of co-processed fuels where RFNBOs is partially replacing a conventional input, a differentiation in the GHG calculation shall be made based on the energetic value of the inputs. In this case, the GHG values may be calculated as an

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average for the entire production of fuels occurring during a period of at most one calendar month.

Within ISCC PLUS, it is not mandatory to meet the 70% GHG savings threshold as under ISCC EU. However, if a system user wishes to provide the emissions savings, it should be calculated as below:

$$Emission\ savings = \frac{(E_F - E)}{E_F}$$

where

E = total emissions from the use of RFNBO

E_F = total emissions from fossil fuel comparator (94 CO_2eq/MJ)

3.1 Emissions from elastic inputs (e_i elastic)

Elastic inputs are those, where the supply can be increased to meet the extra demand. H_2, CO_2, natural gas and electricity can be considered as elastic inputs. Emissions from elastic inputs that are obtained in an incorporated process shall be calculated based on data of their actual production. Emissions arising from the input production over the whole supply chain including extraction of primary energy required to make the input, processing and transport emissions are counted.

For post-industrial and atmospheric CO_2, the supplier of CO_2 (issuer of the sustainability declaration) only provides the effort for the capture and (if applicable) liquefaction of CO_2. The supplier of CO_2 does not calculate GHG emissions. However, the recipient of the CO_2 bears the responsibility to calculate GHG emissions based on the information provided for capture, liquefaction and transport.

When elastic inputs are not obtained from an incorporated process, emission factors can be used from them database such as ECOINVENT, IPCC, IEA, governmental bodies or peer-reviewed publications can be used.

3.1.1 Emission Factors for Electricity

In case the renewable electricity is produced at a respective renewable energy unit and directly consumed by the installation that is producing the certified product, the system user can attribute zero greenhouse gas emissions from the use of renewable electricity. The following requirements shall be fulfilled for this approach:

- The renewable electricity production unit is not connected to the grid or is connected to the grid, but evidence can be provided that the electricity concerned has been supplied without taking electricity from the grid

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> The renewable electricity production unit comes into operation after, or at the same time as, the installation producing the certified product.

In case electricity is supplied by the grid, the emission factors of the national/country electricity mix shall be used[7]. The only exception to use an updated emission factor for electricity is, if the system user has Power Purchase Agreement (PPA) combined with Energy Attribute Certificates (e.g., Guarantees of Origin (GoO) in the EU or RECs in the U.S., I-RECs) or comparable proof of no multiple counting, the system user can choose to apply the supplier specific EF. The following information shall be a part of the documentation for using supplier specific EF:

> The energy source from which the energy carrier was produced and the start and end dates of production.

> The type of energy or energy carrier (e.g., electricity, gas–including H_2, heat, or cooling).

> The identity, location, type and capacity of the installation where the energy or energy carrier was produced.

> The date on which the installation became operational (i.e., starting point of first electricity supply into grid).

> The date and country of issue and a unique identification number of the EAC. (In other words, a proof of no double counting is required).

> The proof of renewability must ensure that double counting of the renewable electricity is avoided. A respective proof of the unit using the renewable energy must be provided during the audit.

If those requirements are not fulfilled, or, in case the supplier specific EF is not available, or contracts are sold to a third party, the emission factors of the national/country electricity mix shall be used.

3.2 Emissions from rigid inputs (e_i rigid)

Rigid inputs are classified as those supply that cannot be expanded to meet the demands. Rigid inputs include liquid or solid waste streams of non-renewable origin which are not suitable for material recovery or waste processing gas and exhaust gas of non-renewable origin. According to EU 2023/1185, all inputs qualifying as a carbon source for the production of recycled carbon fuels are rigid, as well as outputs produced in fixed ratio by an incorporated process[8] and which represent less than 10 % of the economic

[7] The primary source for official national and regional grid emission factors.
[8] Incorporated processes include processes that take place in the same industrial complex, or that supply the input via a dedicated supply infrastructure, or that supply more than half of the energy of all inputs to the production of the renewable liquid and gaseous transport fuel of non-biological origin or recycled carbon fuel.

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value of the output. If it represents 10 % or more of the economic value, it shall be treated as elastic.

The calculation emissions from rigid inputs shall include the emissions from the diversion of those inputs from a previous or alternative use. Loss of production of electricity, heat or products that were previously generated using the input, and any additional emissions originating to the additional treatment of the input and transport shall be included in calculation.

Emissions from the supply of rigid inputs shall be determined by multiplying the lost production of electricity, heat of other material with the relevant emission factor.

3.3 Emissions from existing use or fate (e_{ex-use})

These inputs include avoided emissions such as CO_2 equivalent emissions that would have been emitted if the carbon incorporated in the chemical composition of the fuel would not have been used for fuel production. Inputs such as electricity, heat and consumable materials used in the process of capturing, transporting and storing CO_2 should be accounted in the calculation.

In case CO is used as an input, it should be considered any other input that has an energy content.

3.4 Emissions from Processing (e_p)

It should include all direct atmospheric emissions originating from the processing including combustion, waste treatment and leakages. Hydrogen leakage is considered as an energy loss, with proportional increase of the emissions intensity. The use of processing inputs resulting in direct atmospheric emissions should be accounted for under emissions from processing.

3.5 Emissions from Transport and Distribution (e_{td})

This includes the emissions from storage and transport of the final fuels. Storage and transport related emissions of inputs shall be included in the emissions attributed to the inputs (e_i).

3.6 Emissions from Fuel in Use (e_u)

Emissions from the fuel in use refers to the total combustion emissions of the fuel in use. The greenhouse gases to be considered are CO_2, N_2O and CH_4.

If this approach is followed, this must be made clear for the downstream supply chain with an additional statement in the ISCC PLUS sustainability declaration, in the field “product(s) related information” in annex:

“This GHG value, calculated based on the ISCC EU 205-1 methodology, does not account for the combustion emissions (e_u) as the material is not designated to be used as fuel. If this product is to be used by the recipient for energetic

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purposes (e.g. as fuel for transport, heating and cooling), it is mandatory to add the combustion emissions."

3.7 Emissions saving from carbon capture and geological storage (e_{ccs})

If carbon emissions from the RFNBOs fuel production are permanently stored, these emissions can be considered as e_{ccs}. The requirements for the geological storage of CO_2 should be in accordance with or equivalent to the Directive 2009/31/EC. However, emissions from storage and transport of CO_2 must be considered under e_p.

3.8 Life Cycle Coverage

Under ISCC PLUS, the GHG emissions calculation can either cover the whole life cycle of the product (cradle-to-grave), or only the emissions up to the factory gate (cradle-to-gate).

It must always be clearly highlighted on the Sustainability Declaration of the product if cradle-to-grave or cradle-to-gate approach is used. If required, further information on the additional emissions to be included for the product must be provided to the recipient of the material.

If cradle-to-gate approach is applied for the product, emissions from fuel in use (e_u) need not be considered for the emissions calculation. However, it should be noted that emissions for existing use or fate ($e_{ex\text{-}use}$) cannot be applied when omitting e_u from calculation. Hence, $e_{ex\text{-}use}$ can only be utilised when applying e_u.

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